

Lab 3: Data-Centric Machine Learning with Scikit-Learn

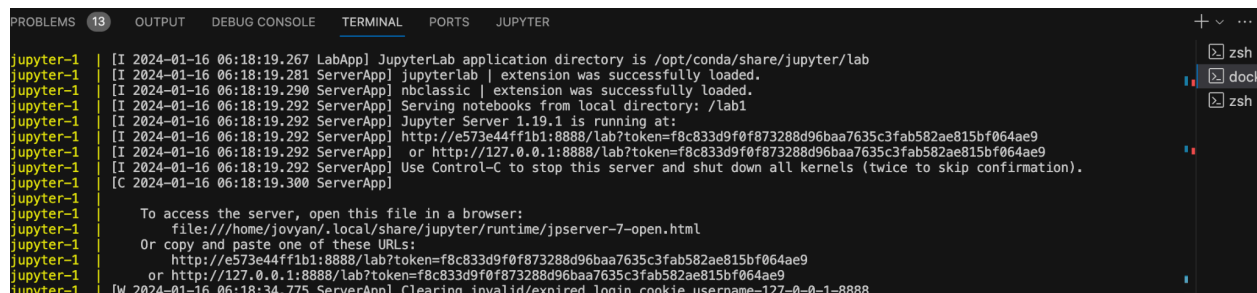
In this lab, you will be using Scikit-Learn to explore various aspects of data centric machine learning, including classification, clustering, dimensionality reduction, and regression. Below are the instructions to get started with the lab.

Setup Instructions Docker

We will be using a docker environment similar to the one which we used in lab 1. Download the lab folder on your local machine and from the lab directory, run this command (where your docker-compose.yml file is located):

```
docker-compose up --build
```

After running docker compose command you will see a url as shown in the image below. Paste the url in the web browser or paste it after selecting jupyter kernel in vscode to open the jupyter notebook.



```
jupyter-1 | [I 2024-01-16 06:18:19.267 LabApp] JupyterLab application directory is /opt/conda/share/jupyter/lab
jupyter-1 | [I 2024-01-16 06:18:19.281 ServerApp] jupyterlab | extension was successfully loaded.
jupyter-1 | [I 2024-01-16 06:18:19.290 ServerApp] nbclassic | extension was successfully loaded.
jupyter-1 | [I 2024-01-16 06:18:19.292 ServerApp] Serving notebooks from local directory: /lab1
jupyter-1 | [I 2024-01-16 06:18:19.292 ServerApp] Jupyter Server 1.19.1 is running at:
jupyter-1 | [I 2024-01-16 06:18:19.292 ServerApp] http://e573e44ff1b1:8888/lab?token=f8c833d9f0f873288d96baa7635c3fab582ae815bf064ae9
jupyter-1 | [I 2024-01-16 06:18:19.292 ServerApp] or http://127.0.0.1:8888/lab?token=f8c833d9f0f873288d96baa7635c3fab582ae815bf064ae9
jupyter-1 | [I 2024-01-16 06:18:19.292 ServerApp] Use Control-C to stop this server and shut down all kernels (twice to skip confirmation).
jupyter-1 | [C 2024-01-16 06:18:19.300 ServerApp]
jupyter-1 |
jupyter-1 | To access the server, open this file in a browser:
jupyter-1 | file:///home/jovyan/.local/share/jupyter/runtime/jpserver-7-open.html
jupyter-1 | Or copy and paste one of these URLs:
jupyter-1 | http://e573e44ff1b1:8888/lab?token=f8c833d9f0f873288d96baa7635c3fab582ae815bf064ae9
jupyter-1 | or http://127.0.0.1:8888/lab?token=f8c833d9f0f873288d96baa7635c3fab582ae815bf064ae9
jupyter-1 | [W 2024-01-16 06:18:34.775 ServerApp] Clearing invalid/expired login cookie username-127-0-0-1-8888
```

There are two notebooks in which you have to work on and submit named,

1. lab3_regression.ipynb
2. lab3_classification.ipynb

There are two tutorial notebook as well which you can refer

1. tutorial_regression.ipynb
2. tutorial_classification.ipynb

Setup Instructions without Docker (Not Recommended)

If you are facing issues with docker installation or setup, then you can use any other environment setup which you already have like Conda or Python, you would have to install the necessary libraries which are imported in the notebook.